

Phase II – Business Process Modeling

Project: Umurenge SACCO Savings & Loan Management System

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1. System Scope

The Umurenge SACCO Savings & Loan Management System automates the entire lifecycle of member savings and loan operations for a local SACCO. The system handles member registration, savings deposits, loan applications, interest calculation, repayment tracking, and financial reporting. It ensures that loan policies are enforced automatically (e.g., blocking new loans for overdue members) and provides real-time visibility into member balances and loan statuses.

2. Actors

- **Member:** An individual who saves money with the SACCO and borrows loans. The member can register, deposit savings, apply for loans, make repayments, and check their balance.
- **Loan Officer:** A SACCO staff member responsible for verifying member information, checking overdue status, approving/rejecting loans, recording repayments, and generating reports.
- **System (Database):** The Oracle database that automatically calculates interest, updates loan balances, blocks overdue members from taking new loans, and generates financial summaries.

3. Workflow Description (Start to End)

The process begins when a member registers with the SACCO and provides their personal details (National ID, name, phone, location). Once registered, the member can make savings deposits, which are recorded in the system and contribute to their total savings balance.

When a member needs a loan, they apply through the loan officer. The loan officer verifies the member's identity and checks the system to see if the member has any overdue loans (loans past their due date by more than 30 days). If the member has overdue loans, the system automatically blocks the application and rejects it. If the member has no overdue loans, the loan officer approves the application.

Upon approval, the system automatically calculates the interest based on the loan type and duration, determines the total amount due, and creates the loan record. The member then begins making repayments (weekly or monthly). Each time a repayment is made, the system updates the remaining balance. Once the balance reaches zero, the system automatically changes the loan status to "PAID."

Finally, the system generates daily and weekly financial reports that summarize total savings, active loans, overdue loans, and member activity. These reports help SACCO administrators make informed decisions.

4. Business Rules Enforced

- A member with an overdue loan (more than 30 days past due date) cannot take a new loan.
- Interest is calculated automatically using the formula: $(\text{Principal} \times \text{Rate} \times \text{Duration}) / 1200$.
- Transactions are blocked on weekends (Saturday & Sunday) and public holidays.
- All changes to loan records are audited and logged for transparency.

5. Expected Benefits

- Eliminates manual errors in interest calculation.
- Enforces loan policies automatically (no human bias).
- Saves staff hours by automating daily reports.
- Improves member satisfaction through faster service.
- Provides complete audit trail for accountability.